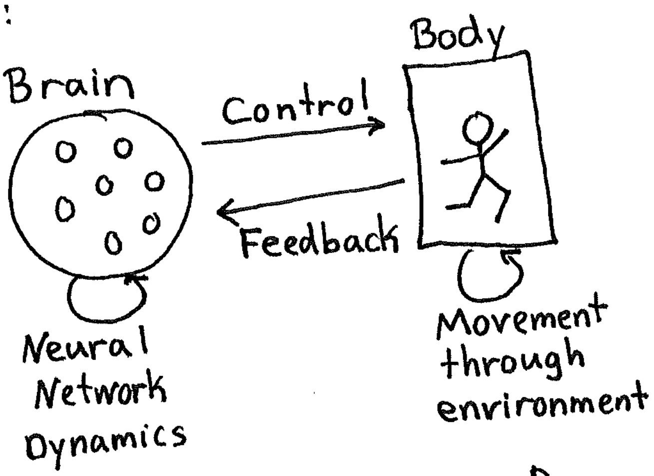
Introduction to motor systems neuroscience:

The brain and body are coupled dynamical systems:

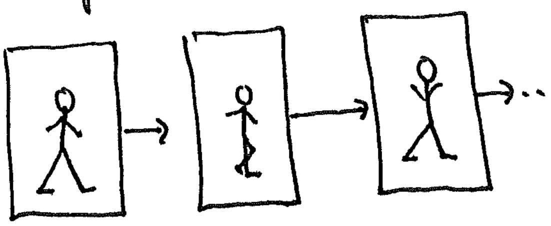


- Feedforward control: neglect feedback
- Feedback control: incorporate feedback (e.g. proprioception)

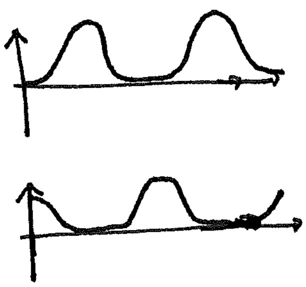
How do the dynamics of recurrent neural networks relate to the dynamics of the body?

Example:

Rhythmic motor output (e.g. walking)



Central pattern generator (e.g., oscillation)



Brain regions involved in motor control

- spinal cord : low-level muscle control
- brainstem
- cerebellum
- basal ganglia
- motor cortex : high-level control of skilled movements

A major challenge is that motor signals are highly distributed across brain. We must understa collective patterns amongst many neurons.

Analyzing recurrence in linear neural networks

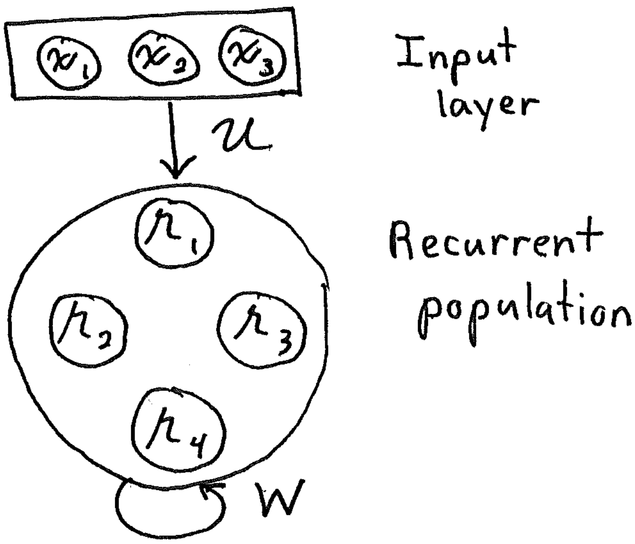
Familiar differential equation:

$$\tau \frac{dr}{dt} = -r + I$$

We now reduce linear neural networks to equations of this form. In general, we derived

$$\underbrace{\tau \frac{dr_i}{dt}}_{\substack{N \text{ coupled} \\ \text{differential} \\ \text{equations}}} = -r_i + \underbrace{f}_{\substack{\text{Firing rate} \\ \text{nonlinearity}}}\Big(\underbrace{\textstyle\sum_{j=1}^{N} W_{ij} r_j}_{\substack{\text{recurrent} \\ \text{input}}} + \underbrace{\textstyle\sum_{j=1}^{M} U_{ij} x_j}_{\text{external input}} \Big)$$

Diagram:



We now specialize to the case where $f(x) = x$

$$\tau \frac{dr_i}{dt} = -r_i + \sum_{j=1}^{N} W_{ij} r_j + I_i$$

where

$$I_i(t) = \sum_{j=1}^{M} U_{ij}\, x_j(t)$$

summarizes the total effect of the input layer on the i^{th} recurrent neuron.

Importantly, the general linear neural network equations can be put in decoupled form by looking at eigenvectors of W. Let's rewrite coupled equations in matrix form.

$$\tau \frac{dr_i}{dt} = -r_i + \sum_{j=1}^{N} W_{ij} r_j + I_i$$

$$\tau \frac{dr}{dt} = -r + Wr + I$$

Let $\xi^{(i)}, \lambda_i$ be eigenvectors and eigenvalues of W.

$$W \xi^{(i)} = \lambda_i \xi^{(i)}$$

Collecting eigenvectors and eigenvalues in matrices

$$V = \left[\xi^{(1)} \; \xi^{(2)} \ldots \xi^{(N)} \right] ; \quad D = \begin{pmatrix} \lambda_1 & & 0 \\ & \lambda_2 \ddots & \\ 0 & & \lambda_N \end{pmatrix}$$

this says

$$WV = VD \Rightarrow V^{-1} W V = D \quad \text{(diagonalization)}$$

$$\text{OR} \quad W = V D V^{-1}$$

This puts firing rate equations into convenient form

$$\underbrace{\tau \frac{dr}{dt} = -r + V D V^{-1} r + I}_{\text{Multiply by } V^{-1} \text{ on left}}$$

$$\tau \left(V^{-1} \frac{dr}{dt} \right) = -\underbrace{(V^{-1} r)}_{\alpha} + D(V^{-1} r) + \underbrace{(V^{-1} I)}_{\beta}$$

$$\tau \frac{d\alpha}{dt} = -\alpha + D\alpha + \beta$$

$$\tau \frac{d\alpha_i}{dt} = -\alpha_i + \lambda_i \alpha_i + \beta_i$$

The α_i variables behave like decoupled neurons wi self-couplings λ_i. Assuming $\lambda_i < 1$, the solution is found through the familiar procedure.

$$\tau \frac{d\alpha_i}{dt} = -(1-\lambda_i)\alpha_i + \beta_i$$

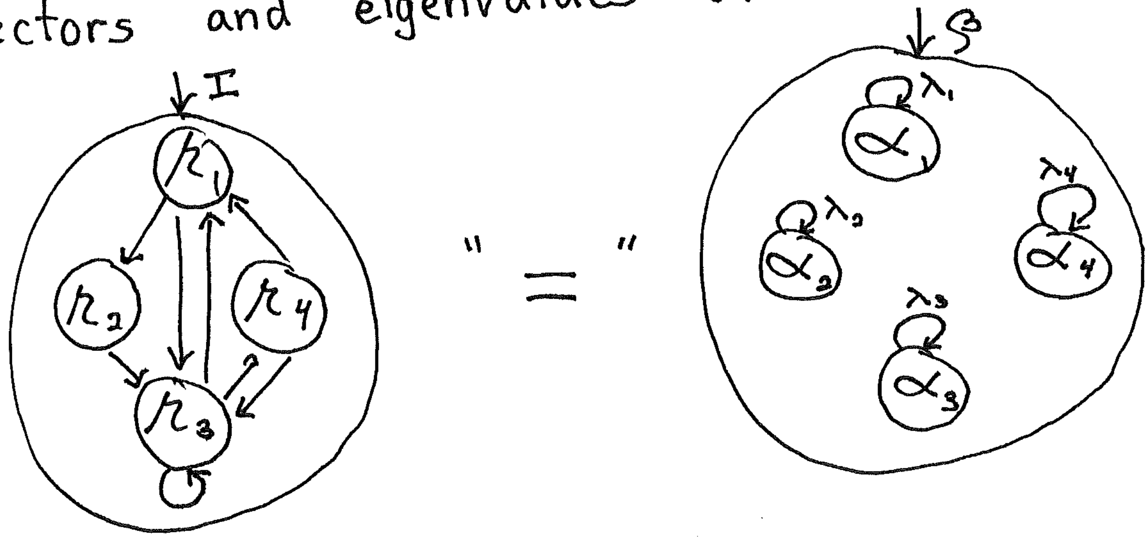
$$\underbrace{\frac{\tau}{1-\lambda_i}}_{\tau_{\text{eff},i}} \frac{d\alpha_i}{dt} = -\alpha_i + \underbrace{\frac{\beta_i}{1-\lambda_i}}_{\beta_{\text{eff},i}}$$

$$\alpha_i(t) = \int_{-\infty}^{t} dt' \, \frac{1}{\tau_{\text{eff},i}} e^{-(t-t')/\tau_{\text{eff},i}} \beta_{\text{eff},i}(t')$$

From this, we back infer the firing rates

$$\alpha = V^{-1} r \Rightarrow r(t) = V\alpha(t)$$

We have a complete solution in terms of eigenvectors and eigenvalues of W.

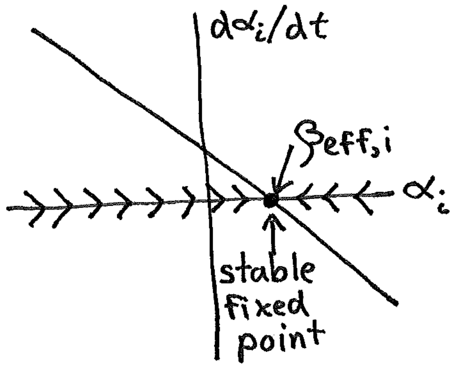


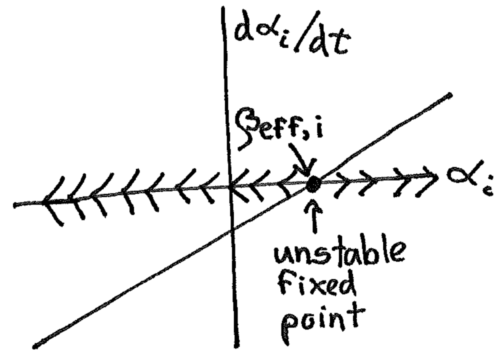
Our current focus on dynamics warrants unpacking this in conceptual terms. Recall

$$\tau_{\text{eff},i} = \frac{\tau}{1-\lambda_i}$$

If $\lambda_i < 1$, then exponential approach dynamics pull α towards β. However, $\tau_{\text{eff},i} < 0$ if $\lambda_i > 1$, leading to exponential growth. A positive feedback loop causes α_i to blow up.

Local decay ($\lambda_i < 1$) Local growth ($\lambda_i > 1$)





Although there is a fixed point with $d\alpha_i/dt = 0$ in bot cases, it's unstable when $\lambda_i > 1$ because small deviations away from it grow with time.

If $\lambda_i = 1$, then the differential equation becomes

$$\tau \frac{d\alpha_i}{dt} = \beta_i \Rightarrow \alpha_i(t) = \alpha_i(0) + \int_0^t dt' \frac{1}{\tau} \beta_i(t')$$

Here "mode" i is a perfect integrator of its inputs because it provides positive feedback that cancels $-\alpha_i$

If λ_i is less than one, but close to it, then $\tau_{eff,i}$ is large and α_i is an approximate integrator of its inputs. It's also called a leaky integrator because recent inputs matter most and distant ones leak awa